

Computational Study of Cancer-associated Variant in the MUTYH–PCNA–DNA Complex

MUTYH is a DNA glycosylase involved in base-excision repair, a pathway that protects genomic stability by removing damaged or incorrectly paired bases from DNA. In particular, MUTYH helps prevent mutations caused by oxidative DNA damage by recognizing and removing adenine incorrectly paired with 8-oxoguanine. Proliferating cell nuclear antigen (PCNA) functions as a sliding clamp that organizes and coordinates proteins involved in DNA replication and repair. The interaction between MUTYH, PCNA, and DNA is therefore important for efficient damage recognition and repair. Because protein function depends not only on static structure but also on molecular motion, amino-acid substitutions may affect repair activity by altering structural stability, flexibility, or communication between different regions of the complex. The R241Q variant has been associated with colorectal cancer, but its potential effects on the dynamics of the MUTYH–PCNA–DNA complex remain unclear. In this study, molecular dynamics simulations were performed using AMBER to compare wild-type (WT) and R241Q complexes in systems containing either a full-length or truncated DNA model. Four conditions were examined: WT and R241Q with full-length DNA and with shortened DNA. Two independent simulation replicates were analyzed for each condition, producing eight trajectories in total. Structural deviation was evaluated using root-mean-square deviation (RMSD) calculations based on protein C α atoms and selected DNA backbone atoms, including P, O3', O5', C3', C4', and C5'. An additional RMSD analysis excluded selected flexible regions of the system to reduce their influence on the overall measurement. Residue-level flexibility was evaluated using root-mean-square fluctuation (RMSF), while coordinated and opposing motions between residues were examined using residue-based dynamic cross-correlation matrices. Across the simulations, the R241Q systems displayed altered RMSD behavior relative to their corresponding WT models, although the magnitude and time dependence of these changes varied between replicates. RMSF profiles showed that the largest fluctuations were concentrated within specific protein regions and at the protein–DNA portions of the system rather than being distributed uniformly throughout the complex. The correlation matrices further indicated that both the R241Q substitution and DNA truncation modified patterns of coordinated and opposing motion between structural regions. Differences between replicate trajectories suggest that the complex samples multiple conformational states and that additional simulations or quantitative statistical comparisons would be required to establish the reproducibility of individual effects. These findings provide a molecular-level basis for understanding how the R241Q substitution may alter DNA-repair complex behavior and may help guide future experimental studies of its potential functional and disease-related consequences.